Advanced Statistics

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Hypothesis Testing using P-value

Instead of using Z/t-statistics in decision rule we can also use p value.

P-value quantifies the probability of observing the test statistic or something more extreme, assuming that $\underline{\underline{H_0}}$ is true.

One tailed test: (Right / left tailed test)

Right tailed : $Z > Z_\alpha \Rightarrow$ Accept H_1 or Reject H_0
Left tailed : $Z < -Z_\alpha \Rightarrow$ Accept H_1 or Reject H_0

$$p \text{ value} = 1 - cdf(z)$$

Two-tailed test : $|z| > Z_{\frac{\alpha}{2}} \Rightarrow$ Accept H_1 or Reject H_0

$$p \text{ value} = 2 \times [1 - cdf(z)]$$

Decision rule using p-value is same for both

p-value $\leq \alpha$: Accept H_1 or Reject H_0
p-value $> \alpha$: Don't accept H_1 or Failed to reject H_0

Categories of statistical Tests

1) Univariate Test

- used to analyze one variable at a time

eg. Chi-square test for goodness of fit, one sample t-test, Kolmogorov-Smirnov test.

2) Bivariate test

- used to evaluate relationships or differences between two variables
- eg. Pearson Correlation test, Chi-square test of independence, Two sample t-test, Spearman correlation test.

3) Multivariate Test

- used to compare three or more groups or variables

eg. One way ANOVA, Kruskal-wallis test